

Single core Copper & Aluminum Conductors

XLPE Insulated, Aluminum Tape armored and PVC Sheathed

Voltage Grade

18 / 30 (36) KV

Description

Soft annealed Stranded circular compacted Copper or Stranded circular compacted Aluminum conductor, semiconducting layer as conductor screen, XLPE insulated semiconducting layer as non metallic insulation screen, Copper tape or wire as metallic insulation screen, armored with aluminum tape and PVC sheathed.

Application

These cables are generally suitable for direct burial or for installation on trays or in ducts.

Technical Data

Relevant Standard	IEC 60502
Conductor	Plain copper conductor class 2 or Aluminum conductor class 2
Conductor Screen	Extruded Semiconductive Compound
Insulation	Extruded XLPE Compound Rated 90° C
Insulation Screen	Non Metallic part: Extruded Semiconductive Compound (Bonded or Strippable) to the insulation Metallic part: (by one of the following) • Copper tape is applied helically with a suitable overlap. • Copper Wires helically applied and banded with a Copper tape to achieve electrical contact • Overlapped Copper Tape then Copper Wires are also available (in single core only) Conductor screen, XLPE insulation and non metallic insulation screen are applied at the same time using triple head extruder.
Inner Sheath	Extruded PVC Compound
Armoring	Double Aluminum Tapes applied helically over the inner sheath.
Outer Sheath	Extruded PVC Compound
Outer Sheath Color	Black (default color), or any other color according to customer request
Maximum Operating Temp.	90° C
Minimum bending Radius	12 x outer diameter
Packing Condition	Our standard length on Wooden Drum / or according to customer request



MEDIUM VOLTAGE CABLES

Product Code	Nominal Cross Sectional Area	Max. Conductor Resistance			Operating Capacitance	Inductance		Current Rating				Approx. Overall Diameter	Approx. Weight
		DC at 20 °C	AC at 90° C			Flat  Trefoil 		Laid in Ground		Laid in Free Air			
								Flat 	Trefoil 	Flat Touched 	Trefoil 		
mm²	Ω/km	Ω/km	Ω/km	μF/km	mH/km	mH/km	A	A	A	A	mm	kg/km	
A - Copper Conductors													
PM-CT14-X9-01-DP	50	0.3870	0.4937	0.4937	0.1436	0.5366	0.4904	224	217	245	241	35.2	1600
PM-CT15-X9-01-DP	70	0.2680	0.3420	0.3420	0.1615	0.5081	0.4618	276	266	303	301	37.2	1885
PM-CT16-X9-01-DP	95	0.1930	0.2465	0.2466	0.1762	0.4865	0.4402	328	315	366	365	38.7	2175
PM-CT17-X9-01-DP	120	0.1530	0.1956	0.1957	0.1898	0.4696	0.4234	372	360	424	420	40.5	2490
PM-CT18-X9-01-DP	150	0.1240	0.1588	0.1589	0.2042	0.4555	0.4093	410	402	481	474	42.2	2905
PM-CT19-X9-01-DP	185	0.0991	0.1273	0.1274	0.2215	0.4400	0.3937	461	454	551	545	44.0	3310
PM-CT20-X9-01-DP	240	0.0754	0.0973	0.0975	0.2434	0.4228	0.3765	530	528	649	642	46.5	3940
PM-CT21-X9-01-DP	300	0.0601	0.0782	0.0784	0.2643	0.4101	0.3639	588	588	742	735	48.9	4610
PM-CT22-X9-01-DP	400	0.0470	0.0619	0.0624	0.2898	0.3969	0.3507	657	658	844	838	51.8	5615
PM-CT23-X9-01-DP	500	0.0366	0.0494	0.0499	0.3209	0.3834	0.3372	735	734	973	954	55.3	6780
PM-CT24-X9-01-DP	630	0.0283	0.0395	0.0403	0.3594	0.3735	0.3273	810	829	1090	1097	59.8	8255
PM-CT25-X9-01-DP	800	0.0221	0.0325	0.0335	0.4006	0.3605	0.3142	884	912	1222	1234	64.2	10165
PM-CM26-X9-01-DP	1000	0.0176	0.0249	0.0252	0.4921	0.3581	0.3119	1062	1093	1507	1519	75.4	13405
B - Aluminum Conductors													
PM-AT14-X9-01-DP	50	0.6410	0.8219	0.8219	0.1476	0.5291	0.4829	173	169	191	188	35.6	1340
PM-AT15-X9-01-DP	70	0.4430	0.5682	0.5682	0.1624	0.5055	0.4593	212	207	236	234	37.3	1480
PM-AT16-X9-01-DP	95	0.3200	0.4105	0.4106	0.1781	0.4840	0.4378	254	247	287	284	38.9	1620
PM-AT17-X9-01-DP	120	0.2530	0.3247	0.3248	0.1907	0.4685	0.4223	288	281	332	326	40.6	1785
PM-AT18-X9-01-DP	150	0.2060	0.2645	0.2646	0.2071	0.4527	0.4065	320	315	375	370	42.5	2035
PM-AT19-X9-01-DP	185	0.1640	0.2108	0.2109	0.2244	0.4376	0.3914	361	358	432	428	44.3	2220
PM-AT20-X9-01-DP	240	0.1250	0.1610	0.1612	0.2434	0.4228	0.3765	417	414	509	503	46.5	2485
PM-AT21-X9-01-DP	300	0.1000	0.1291	0.1293	0.2643	0.4101	0.3639	461	465	585	577	48.9	2765
PM-AT22-X9-01-DP	400	0.0778	0.1010	0.1012	0.2898	0.3969	0.3507	528	530	679	673	51.8	3250
PM-AT23-X9-01-DP	500	0.0605	0.0793	0.0797	0.3209	0.3834	0.3372	596	602	779	776	55.3	3730
PM-AT24-X9-01-DP	630	0.0469	0.0623	0.0628	0.3604	0.3732	0.3270	672	683	908	900	59.9	4390
PM-AT25-X9-01-DP	800	0.0367	0.0500	0.0507	0.4006	0.3605	0.3142	751	767	1030	1033	64.2	5100
PM-AM26-X9-01-DP	1000	0.0291	0.0389	0.0391	0.4921	0.3581	0.3119	905	919	1277	1274	75.4	6775

The above data is approximate and subject to manufacturing tolerances

Note:-

Flame retardant PVC sheathed or Polyethylene Sheathed are available according to customer request
Longitudinal or radial Water tightness layers are also available according to customer request
the below data are applicable for cables with rated voltage 19/33 (36) KV